

# JavaScript Loop and Pattern Problems

## Question 1: Reverse a Number

Input: 12345

Output: 54321

Explanation: Extract the last digit using % 10, add it to a new number in reverse order, and remove the last digit using Math.floor(number/10).

## Question 2: Sum of Digits

Input: 12345

Output: 15

Explanation: Keep extracting the last digit and add it to a running sum until the number becomes 0.

## Question 3: Count Digits

Input: 12345

Output: 5

Explanation: Count how many times you can remove the last digit before the number becomes 0.

## Question 4: Palindrome Number

Input: 121

Output: true

Explanation: Reverse the number and compare it with the original. If both are equal, it is a palindrome.

## Question 5: Armstrong Number

Input: 153

Output: true

Explanation: Count digits, raise each digit to the power of total digits, add them, and compare the sum with the original number.

## Question 6: Factorial

Input: 5

Output: 120

Explanation: Multiply numbers from 5 down to 1 using a while loop.

### Question 7: Fibonacci Series

Input: 10 terms

Output: 0 1 1 2 3 5 8 13 21 34

Explanation: Each next number is the sum of the previous two numbers.

### Question 8: Decimal to Binary

Input: 10

Output: 1010

Explanation: Repeatedly divide the number by 2 and store the remainders.

### Question 9: Largest Digit in a Number

Input: 927451

Output: 9

Explanation: Compare each extracted digit with the current maximum digit.

### Question 10: Prime Number Check

Input: 17

Output: true

Explanation: Check whether any number between 2 and n-1 divides the number completely.

## Pattern Problems

### Question 11: Square Star Pattern

Input: n=5

Output:

\* \* \* \* \*

(repeated 5 rows)

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 12: Right Triangle

Input: n=5

Output:

\*

\* \*

\* \* \*

```
* * * *  
* * * * *
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 13: Reverse Triangle

Input: n=5

Output:

```
* * * * *  
* * * *  
* * * *  
* * *  
* *  
*
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 14: Number Triangle

Input: n=5

Output:

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 15: Same Number Triangle

Input: n=5

Output:

```
1  
2 2  
3 3 3  
4 4 4 4  
5 5 5 5 5
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 16: Pyramid

Input: n=5

Output:

```
*  
**  
***
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 17: Inverted Pyramid

Input: n=5

Output:

```
*****  
****  
***  
**  
*
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 18: Diamond

Input: n=5

Output:

Upper Pyramid + Lower Inverted Pyramid

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 19: Floyd's Triangle

Input: n=5

Output:

```
1  
2 3  
4 5 6
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 20: Binary Triangle

Input: n=5

Output:

```
1  
0 1  
0 1 0
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 21: Alphabet Triangle

Input: n=5

Output:

A

A B

A B C

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 22: Reverse Number Triangle

Input: n=5

Output:

5 4 3 2 1

5 4 3 2

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 23: Hollow Square

Input: n=5

Output:

\*\*\*\*\*

\*   \*

\*   \*

\*\*\*\*\*

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 24: Hollow Rectangle

Input: rows=4 cols=6

Output:

\*\*\*\*\*

\*   \*

\*\*\*\*\*

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 25: Butterfly Pattern

Input: n=5

Output:

```
*   *  
**  **
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 26: X Pattern

Input: n=5

Output:

```
* *  
* *  
*  
* *  
* *
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 27: Plus Pattern

Input: n=5

Output:

```
  *  
*****
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 28: Pascal Triangle

Input: n=5

Output:

```
1  
1 1  
1 2 1
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 29: Multiplication Table Pattern

Input: n=5

Output:

```
1 2 3 4 5  
2 4 6 8 10
```

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.

### Question 30: Chess Board Pattern

Input: n=5

Output:

\* # \* # \*

# \* # \* #

Task: Write a JavaScript program using nested for loops to generate the above pattern exactly as shown.